Lecture 4 (February 10)

We'll discuss some generalizations of the Catalan numbers over the coming days.

Young Tableaux

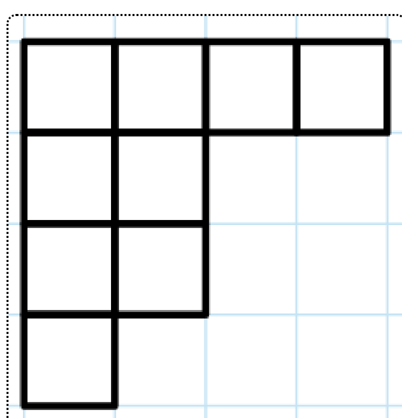
We will consider partitions today. A partition is essentially a way to write an integer as the sum of positive integers. Order doesn't matter for partitions, so by convention we write the parts of a partition in decreasing order. (If we do care about the order, we have a different type of structure called a composition.) More formally:

Definition (Partitions).

We call $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_k)$ a **partition** of n ($\lambda \vdash n$) if $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k \geq 1$ and $\lambda_1 + \lambda_2 + \cdots + \lambda_k = n$.

For example, $3 + 2 = 5$ and $2 + 3 = 5$ both correspond to the partition $(3, 2)$ of 5.

We usually represent partitions graphically using **Young diagrams**, which are shapes made up of boxes. This is best illustrated with an example: for $\lambda = (4, 2, 2, 1) \vdash 9$, we illustrate it with nine boxes in four rows, each corresponding to one part of the partition.



Note how we align the left edges of each row (which are still in decreasing size order). We can also use λ to refer to the diagram (rather than its underlying partition).

> **Definition** (Standard Young Tableaux).
>
> A **standard Young tableau** (SYT; plural: "tableaux") of a shape $\lambda \vdash n$ is a way to fill the boxes of λ with $1, 2, \ldots, n$ (without repetitions) such that the entries increase in the rows and columns of λ.

An example for the above λ is:

1	3	4	6
2	5		
7	9		
8			

We denote the number of SYTs of a shape λ as f_λ (sometimes people use f^λ, but that can be confused for an exponent).

For example, $f_{(3)} = 1$, $f_{(2,1)} = 2$, and $f_{(2,2)} = 2$. (In lecture these were represented with the physical Young diagrams as subscripts, but I cannot typeset them that way.)

Let's compute $f_{(3,2)}$. We know 1 must be in the top left. If 2 is in the bottom left, there are two options to place $3, 4, 5$. If 2 is in the top-middle box, there are three options for $3, 4, 5$. So, in total, $f_{(3,2)} = 2 + 3 = 5$.

$f_{(3,3)}$ is also 5 (6 is forced to be in the bottom right, and then we've reduced the problem to $\lambda = (3, 2)$.)

How about $f_{(4,4)}$? It's harder to list all of the possibilities out, but we can make a guess based on our previous lectures: perhaps it's 14, and perhaps $f_{(5,5)}$ is 42...

Proposition 1 ($f_{(n,n)} = C_n$).

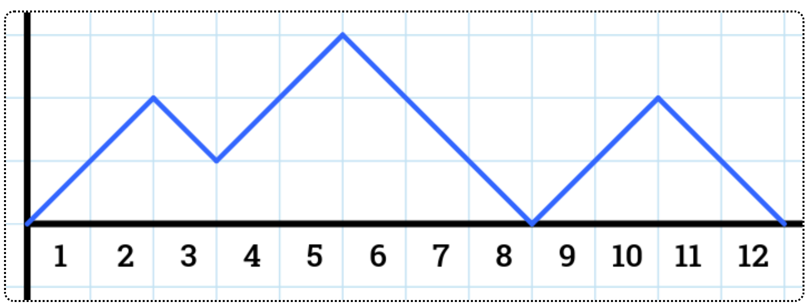
$f_{(n,n)}$, the number of standard Young tableaux for a diagram comprising two rows of n boxes, is C_n.

Proof. We form a bijection between SYTs of shape (n, n) and Dyck paths with $2n$ steps. The numbers in the first row of the tableau correspond to up steps, and the numbers in the second row correspond to down steps.

As an example,

$$T = \begin{array}{|c|c|c|c|c|c|} \hline 1 & 2 & 4 & 5 & 9 & 10 \\ \hline 3 & 6 & 7 & 8 & 11 & 12 \\ \hline \end{array}$$

corresponds to the following Dyck path.



The condition that numbers increase down columns is equivalent to the condition that the Dyck path does not go below the x-axis: it's essentially saying that the kth up step comes before the kth down step.

Note that we can think of any SYT for a partition with two parts as a lattice path in this way: for example, if we just consider the boxes with $1 \ldots 5$ in the above T, we end up with a lattice path to $(5, 3)$ that does not dip below the x-axis.

We can think of general standard Young tableaux as multi-dimensional Dyck paths.

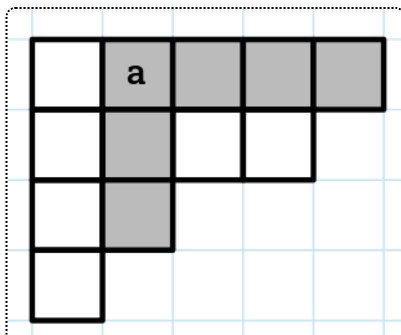
The Hook Length Formula

We would like to find a way to compute f_λ for any λ. Luckily, there is a nice formula!

Definition (Hooks and hook length)**.**

The **hook** of a box a in a Young diagram is the set of boxes that are either directly below or directly to the right of a, along with a itself. The **hook length** $h(a)$ is the number of boxes in the hook of a.

For example, in the below diagram, $h(a) = 6$ and the hook is shaded.



We also write:

$$H(\lambda) = \prod_{a \in \lambda} h(a).$$

Theorem 2 (Hook length formula)**.**

For all partitions $\lambda \vdash n$,

$$f_\lambda = \frac{n!}{H(\lambda)}.$$

Note that it may not even be immediately obvious that $H(\lambda)$ divides $n!$.

This formula was discovered by Frame, Robinson, and Thrall in 1953.

An example: for $(3, 2)$, the hook lengths are $4, 3, 1, 2, 1$ (in reading order), and indeed, we have that

$$f_{(3,2)} = \frac{5!}{1 \cdot 1 \cdot 2 \cdot 3 \cdot 4} = 5.$$

A corollary of the hook length formula is that $C_n = \frac{1}{n+1}\binom{2n}{n}$ (convince yourself of this).

Proof ("Hook walk proof", Greene, Nijenhuis, Wilf 1979)

There are many proofs of the hook length formula; we will present one reasonable one in class. It is a "probabilistic proof" (in the sense that we will use elementary probability theory).

We want to write a recurrence relation for f_λ, so that we can write it in terms of $f_{\lambda'}$ for slightly smaller λ'.

> **Lemma 3** (Recurrence relation).
>
> $$f_\lambda = \sum_{c \text{ is a corner of } \lambda} f_{\lambda \setminus c} \qquad \text{and} \qquad f_\varnothing = 1$$

A "corner" of λ is a box with hook length 1 (i.e. it is the last box in its row and column, and can be removed while preserving the validity of the diagram). $\lambda \setminus c$ is λ after removing c. For example:

$$f_{\yng(3,2)} = f_{\yng(2,2)} + f_{\yng(3,1)}$$

(Indeed, $5 = 2 + 3$).

The proof of this lemma is simple: n must go in one of the corners, and the rest of the tableau is a SYT for $(\lambda \setminus c) \vdash (n-1)$.

So now we need to show that

$$\frac{n!}{H(\lambda)} \stackrel{?}{=} \sum_{c \text{ corner}} \frac{(n-1)!}{H(\lambda \setminus c)}.$$

By dividing out the left-hand side, we see that we need to prove the following:

Theorem 4 (Probability?).

Given a Young diagram λ:

$$\sum_{c \text{ corner}} \frac{1}{n} \frac{H(\lambda)}{H(\lambda \setminus c)} = 1.$$

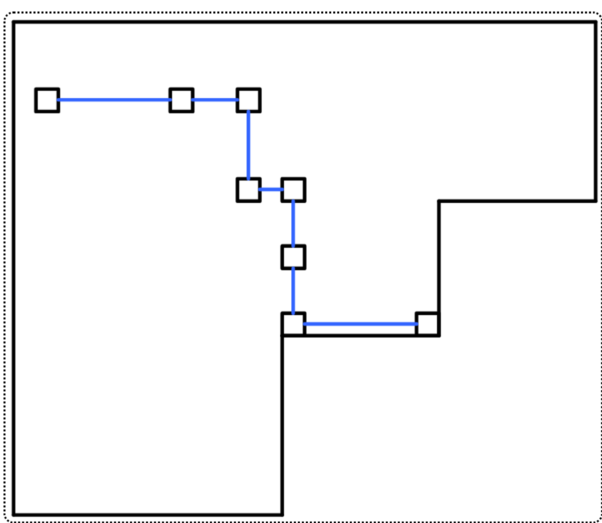
The shape of this equation (a bunch of fractions adding to 1) reminds us of probabilities. Can we come up with a random variable that resolves to each corner c with probability $\frac{1}{n}\frac{H(\lambda)}{H(\lambda\setminus c)}$?

Yes. Greene, Nijenhuis and Wilf did so using the "**hook walk**" algorithm, as follows:

Fix $\lambda \vdash n$.

1. Randomly select a box a of λ.
2. Stop if it is a corner.
3. Jump to any other box in the hook of a (besides a itself), with uniform probability $\frac{1}{h(a)-1}$.
4. Return to step 2.

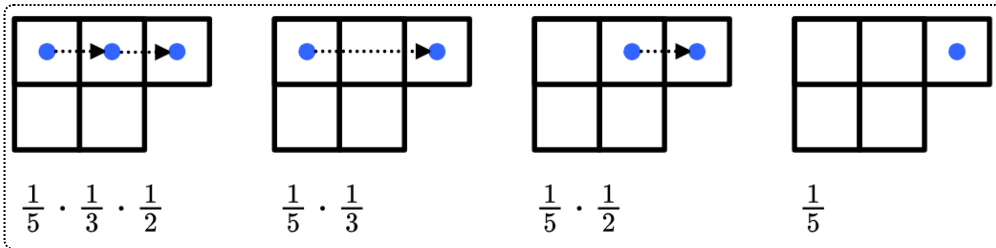
Evidently, this process must terminate and always end up at a corner. This output is our random variable. Here's an example of what the process might look like:



We want to show that we land at a specific corner c with probability $\frac{1}{n}\frac{H(\lambda)}{H(\lambda \backslash c)}$.

For example, let $\lambda = (3, 2)$. There are four hook walks that end in the corner c_1 on the first row. We hope that the probability that we get one of these hook walks is $\frac{1}{n}\frac{H(\lambda)}{H(\lambda \backslash c_1)} = \frac{1}{5}\frac{24}{12} = \frac{2}{5}$.

Indeed:



$$\frac{1}{5} \cdot \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{5} \cdot \frac{1}{3} + \frac{1}{5} \cdot \frac{1}{2} + \frac{1}{5} = \frac{2}{5}$$

We will show that this works in general in the next lecture.

Lecture 5 (February 12)

Today we will complete our proof of the hook length formula.

However, let's first mention a seemingly-very-simple probabilistic proof, as follows:

Claim (Proof?).

Randomly fill up our shape λ with numbers from 1 to n (without any restrictions). Naturally, there are $n!$ ways to do this. We want to show that the probability that it's a valid SYT is $\frac{1}{H(\lambda)}$. Let's just check every box a: it needs to be the largest number in its hook, which has a probability of $\frac{1}{h(a)}$. So the total probability is $\prod \frac{1}{h(a)} = \frac{1}{H(\lambda)}$, and the number of SYTs is $\frac{n!}{H(\lambda)}$. Yay!(?)

This "proof" unfortunately does not show that these probabilities are independent (so the multiplication is invalid), even though it ends up yielding the correct answer. So, just make sure that you're careful.

Continuing the hook walk proof

Scroll up for a recap of what we've done so far. Essentially, we create a random "hook walk" (sequence of boxes) and want to show that the probability of the walk ending at a certain corner c is $\frac{1}{n}\frac{H(\lambda)}{H(\lambda \backslash c)}$, from which we can complete the proof using a recurrence.

For a particular path $\mathcal{P} = (a_1, a_2, \ldots, a_{k-1}, a_k = c)$, the probability that we get that path is

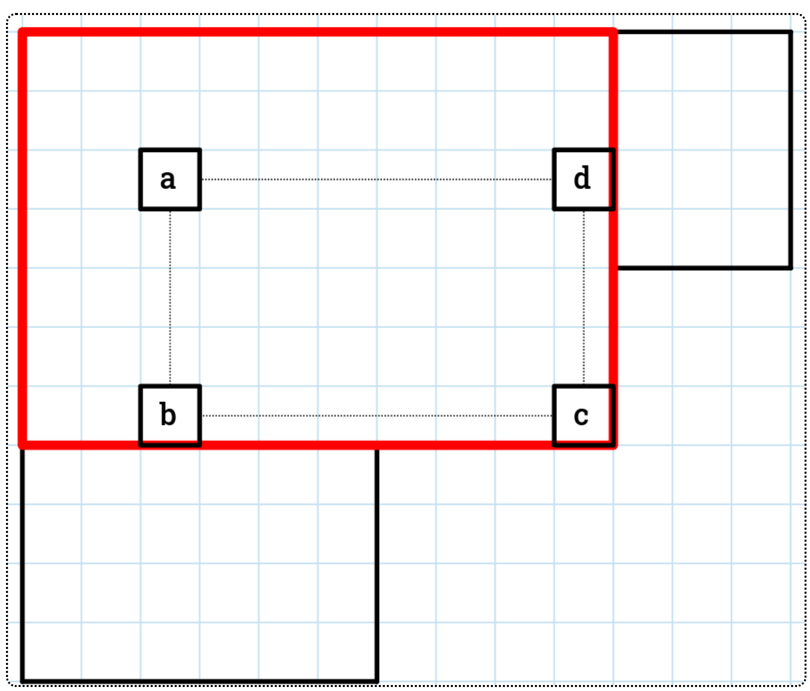
$$\frac{1}{n} \cdot \frac{1}{h(a_1) - 1} \cdot \frac{1}{h(a_2) - 1} \cdots \frac{1}{h(a_{k-1} - 1)}.$$

To simplify this, we define *weights* of boxes and paths in λ. For a non-corner a, we define $\mathrm{wt}(a) = \frac{1}{h(a)-1}$ and for a path $\mathcal{P}$, $\mathrm{wt}(\mathcal{P}) = \mathrm{wt}(a_1) \cdot \mathrm{wt}(a_2) \cdots \mathrm{wt}(a_{k-1})$. So, the hook length formula is equivalent to

$$\sum_{\substack{\text{hook walks} \\ \text{ending at } c}} \mathrm{wt}(\mathcal{P}) = \frac{H(\lambda)}{H(\lambda \setminus c)}.$$

(Note that we removed the $\frac{1}{n}$ term from both sides).

Let's fix λ and a corner c. Note that any hook path $\mathcal{P}$ ending at c must be confined to the rectangle with bottom-right corner at c, as shown in red below:



We want to figure what the $h(a) - 1$ is doing. Observe that with the notation in the above diagram, $h(a) + h(c) = h(b) + h(d)$ and therefore

$$(h(a) - 1) + \cancelto{=0}{(h(c) - 1)} = (h(b) - 1) + (h(d) - 1).$$

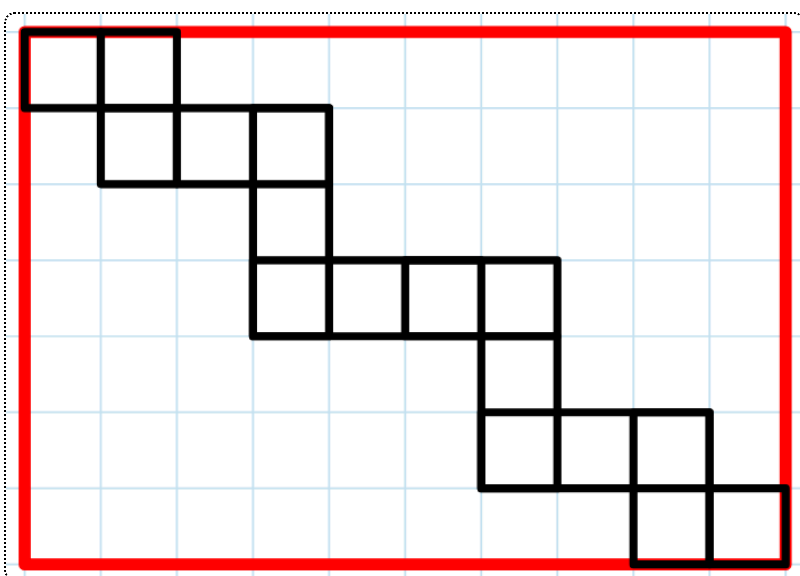
So we only really need to know the hook lengths for boxes in the same row or column as corner c. We use $x_1, x_2, \ldots, x_k$ and $y_1, y_2, \ldots, y_\ell$ to refer to $h(b) - 1$ for all boxes b in the same row/column as corner c.

This way, our weights within the rectangle look like this:

$\frac{1}{x_1+y_1}$	$\frac{1}{x_2+y_1}$	$\cdots\cdots\cdots$	$\frac{1}{x_k+y_1}$	$\frac{1}{y_1}$
$\frac{1}{x_1+y_1}$	$\frac{1}{x_2+y_2}$	$\cdots\cdots\cdots$	$\frac{1}{x_k+y_2}$	$\frac{1}{y_2}$
$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
$\frac{1}{x_1+y_\ell}$	$\frac{1}{x_2+y_\ell}$	$\cdots\cdots\cdots$	$\frac{1}{x_k+y_\ell}$	$\frac{1}{y_\ell}$
$\frac{1}{x_1}$	$\frac{1}{x_2}$	$\cdots\cdots\cdots$	$\frac{1}{x_k}$	1

where c is the bottom-right cell in this grid.

Let's look at lattice paths inside the rectangle from the upper-left to the bottom-right corner (where it can only move one box to the right or downwards). We will see how these are related to hook walks. An example:



Lemma 5 (Total weight of lattice paths).

$$\sum_{\substack{\text{lattice paths}\\ \text{in rectangle}}} \mathrm{wt}(\mathcal{P}) = \frac{1}{x_1 x_2 \dots x_k \cdot y_1 y_2 \dots y_\ell}.$$

For example, if $k = \ell = 1$, there are two lattice paths with weights $\frac{1}{x_1+y_1} \cdot \frac{1}{x_1}$ and $\frac{1}{x_1+y_1} \cdot \frac{1}{y_1}$, which do add up to $\frac{1}{x_1 \cdot y_1}$. (Note that in general, there are $\binom{k+\ell}{k}$ lattice paths, since there are k right steps and ℓ down steps).

Proof of Lemma 5. We induct on $k + \ell$.

Base case: $k = \ell = 0$ works since the empty product is 1. (If you don't like this, you can use $k = \ell = 1$).

Inductive step: We want to subdivide our sum into two smaller sums, based on whether the first step is to the right or downwards. Call these Σ' and Σ'', respectively.

Notice that after the first step, the remainder of the path is just a lattice path in a slightly smaller rectangle, and we know the sum of their weights by induction. So

$$\begin{aligned} \Sigma' &= \frac{1}{x_1 + y_1} \frac{1}{x_2 x_3 \dots x_k \cdot y_1 y_2 \dots y_\ell} \\ \Sigma'' &= \frac{1}{x_1 + y_1} \frac{1}{x_1 x_2 \dots x_k \cdot y_2 y_3 \dots y_\ell}. \end{aligned}$$

(The $\frac{1}{x_1+y_1}$ term is from the first box in all paths, the top-left corner).

So by algebraic simplification we see that our sum is

$$\frac{1}{x_1 + y_1} \left(\frac{1}{x_1} + \frac{1}{y_1} \right) \frac{1}{x_2 x_3 \dots x_k \cdot y_1 y_2 \dots y_\ell} = \frac{1}{x_1 x_2 \dots x_k \cdot y_1 y_2 \dots y_\ell},$$

as desired.

Alright, that's cool, but that's still not the sum of all hook walks (where you can start at any box and jump over any number of boxes as long as you're going right/down). However, we are now able to prove the following:

Lemma 6 (Total weight of hook walks)**.**

$$\sum_{\substack{\text{hook walks}\\ \text{ending at } c}} \mathrm{wt}(\mathcal{P}) = \left(1+\frac{1}{x_1}\right)\left(1+\frac{1}{x_2}\right)\cdots\left(1+\frac{1}{x_k}\right)\left(1+\frac{1}{y_1}\right)\cdots\left(1+\frac{1}{y_\ell}\right)$$

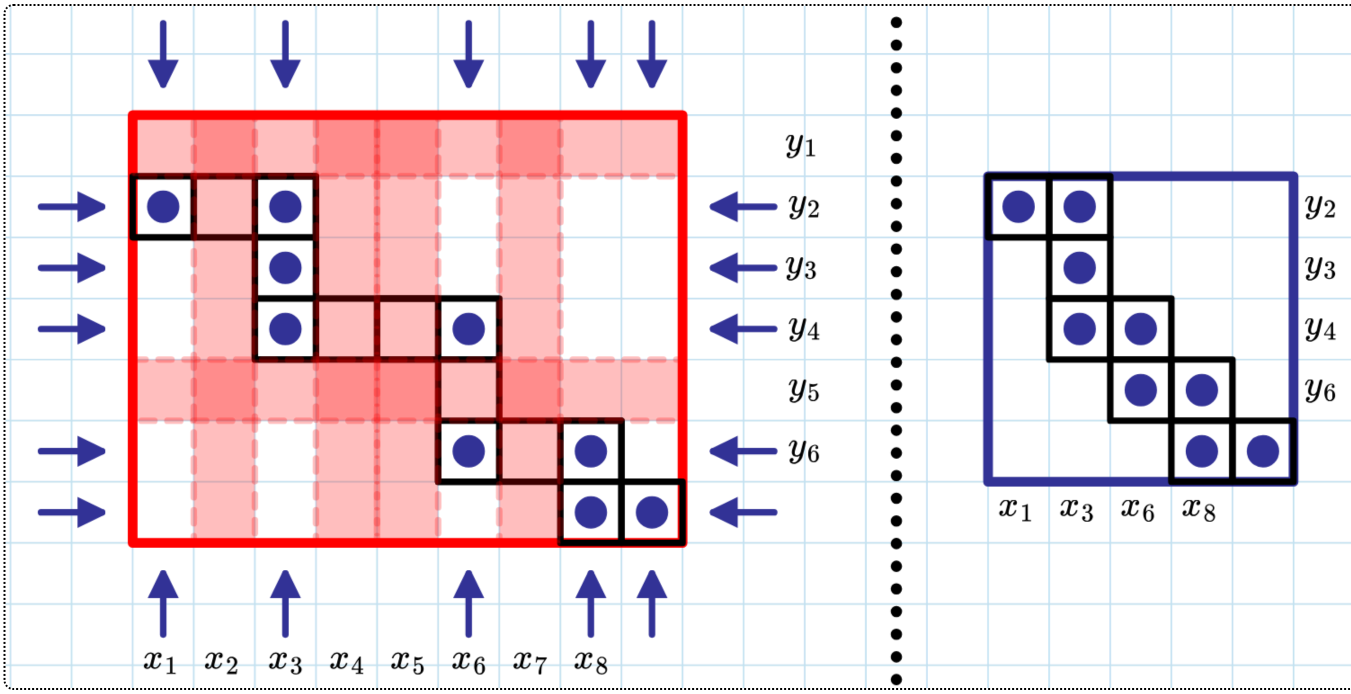
Proof of Lemma 6. Note that we can cleanly relate hook walks to lattice paths in subgrids:

- Each hook walk can be uniquely extended to a lattice path starting from its first box (not necessarily the top-left corner) by re-adding the boxes that were "jumped over".
- If a hook walk jumps over a row/column, it can never go back to that row/column (since hook walks only go right and down).
- As such, if we remove all of the skipped rows/columns (as well as the ones before the hook walk starts), we obtain a subgrid of the original rectangle where the hook walk must be a lattice path.

The reverse correspondence is straightforward: given a lattice path in a subgrid, re-add the other rows/columns, and we have a hook walk.

Here's an example, showing how one particular hook walk corresponds to a subgrid. Note that we carry over the corresponding weights from the original grid, so that the weight of our hook walk is exactly the same as the weight of the lattice path in the subgrid.

The hook walk is represented by dots; the rows/columns with arrows are the ones selected in the subgrid while the ones shaded in red are removed.



So, the sum we want is

$$\sum_{\text{subgrids}} \left(\sum_{\substack{\text{lattice paths} \\ \text{in subgrid}}} \mathrm{wt}(\mathcal{P}) \right).$$

For example, the above subgrid contributes

$$\frac{1}{x_1 x_3 x_6 x_8 \cdot y_2 y_3 y_4 y_6}$$

to this sum by Lemma 5. We can see that the desired expression (in Lemma 6) follows: for any given subgrid, each row term $\frac{1}{y_i}$ or column term $\frac{1}{x_j}$ can be either included or skipped in its product, and each combination of choices corresponds to a valid subgrid.

We can rewrite Lemma 6 as

$$\sum_{\text{hook walks}} \mathrm{wt}(\mathcal{P}) = \prod_{\substack{a \,\in\, \mathrm{cohook}(c) \\ a \neq c}} \left(1 + \frac{1}{h(a) - 1}\right) = \prod_{\substack{a \,\in\, \mathrm{cohook}(c) \\ a \neq c}} \frac{h(a)}{h(a) - 1},$$

where the "cohook" of c is the set of cells directly to the left or directly above c (including c itself; in this case, since c is a corner, it comprises all cells in its row/column).

Q: Why does this product equal $\frac{H(\lambda)}{H(\lambda \setminus c)}$?

A: Look at the contribution of each box a to the product. If it's not in the cohook of c, removing c does not change its hook length, so it doesn't contribute to the product. But if a *is* in the cohook of c (and is not c itself), it contributes a factor of $h(a)$ to $H(\lambda)$ and a factor of $h(a) - 1$ to $H(\lambda \setminus c)$, exactly yielding our factor $\frac{h(a)}{h(a)-1}$.

This concludes the proof of the hook length formula. Yay!